Question	Answer	Marks	Guidance
4(a)	<u>change</u> in internal energy = work done + transfer of thermal energy	C1	**Allow** any equation in which two quantities out of change in internal energy, work done and thermal energy transferred are added/subtracted to give the third. **Allow** ‘heat’ for ‘thermal energy’. **Ignore** symbols unless defined. **Do not allow** <u>change</u> in work / thermal energy. Equality and summation must be clear (not just ‘is’ and ‘and’)
	<u>increase</u> in internal energy = work done <u>on</u> system + energy transferred <u>to</u> the system by heating	A1	**Allow** any set of clearly stated directions for the change in internal energy, work done and thermal energy transferred that is consistent with the version of the equation that is given. **Allow** ‘system’ and ‘gas’ interchangeably. **Ignore** definitions (correct or not) of internal energy. Equality and summation must be clear (not just ‘is’ and ‘and’)
4(b)(i)	$pV = NkT$	C1	**Allow** $pV = nRT$ <u>and</u> $N = nN_A$
	$V = (3.58 \times 10^{24} \times 1.38 \times 10^{-23} \times 298) / (1.84 \times 10^5)$ $= 0.0800\ \text{m}^3$	A1	Full substitution and final answer to 3 significant figures needed. **Allow** ‘$8.31 / 6.02 \times 10^{23}$’ for ‘$1.38 \times 10^{-23}$’. Unit not needed but any unit given must be correct.
4(b)(ii)	$n = (3.58 \times 10^{24}) / (6.02 \times 10^{23})$ $= 5.95$ mol	A1	Correct to at least 3 significant figures. AFC. (5.947) **Allow** AFC for 5.94 mol (from use of value of V in **4(b)(i)** with $pV = nRT$). No ECF from **4(b)(i)**.
4(b)(iii)	$E_K = (3/2)kT$	C1	
	$= (3/2) \times 1.38 \times 10^{-23} \times 298$ $= 6.17 \times 10^{-21}$ J	A1	Correct to at least 3 significant figures. AFC. (6.169)

Question	Answer	Marks	Guidance
4(b)(iv)	no intermolecular forces	B1	**Allow** 'negligible' for 'no'.
4(b)(v)	$U = 3.58 \times 10^{24} \times 6.17 \times 10^{-21}$ $= 2.21 \times 10^{4}$ J	A1	Possible ECF from **4(b)(iii)**. Correct to at least 3 significant figures. AFC. (2.208) **Allow** 22.1 kJ.
4(c)(i)	no change in volume <u>so</u> no work done (on/by gas)	B1	**Allow** in symbols e.g. $\Delta V = 0$ <u>so</u> $W = 0$
	$Q = [(584 - 184) / 184]\ U_A$ $= (+)50\ U_A / 23$ **or** $(+)2.2\ U_A$	A1	Correct to at least 2 significant figures. (2.17). **Allow** 'U' for 'U_A'. **Allow** any simple ratio that computes to 2.2 rounded to 2 significant figures.
4(c)(ii)	(no thermal energy transfer so) increase in internal energy = work done on gas **or** internal energy at C is double the internal energy at A	B1	**Allow** '(no thermal energy transfer so) decrease in internal energy = work done by gas'. **Allow** $U_C = 2U_A$
	$W = U_A - (50\ U_A / 23)$ $= -27\ U_A / 23$ **or** $-1.2\ U_A$	A1	Possible ECF from **4(c)(i)** if Q in terms of U_A only Correct to at least 1 decimal place. (–1.17). **Allow** 'U' for 'U_A'. **Allow** any simple ratio that computes to –1.2 rounded to 2 significant figures.